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Class: 4C

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
```

```
plt.style.use('default')
```

```
df = pd.read_csv('https://drive.google.com/uc?id=1qtRr5cyHF1K9-r-9AJgX-1jvvnhZwGGz')
```

```
print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
df
print("\nDataset Info:")
df.info()
print("\nBasic Statistics:")
df.describe()
```

```
Dataset Shape: (350, 11)

First 5 rows:

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 11 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Customer ID               350 non-null    int64
 1   Gender                    350 non-null    object
 2   Age                       350 non-null    int64
 3   City                      350 non-null    object
 4   Membership Type           350 non-null    object
 5   Total Spend               350 non-null    float64
 6   Items Purchased           350 non-null    int64
 7   Average Rating            350 non-null    float64
 8   Discount Applied          350 non-null    bool
 9   Days Since Last Purchase  350 non-null    int64
 10  Satisfaction Level        350 non-null    object
dtypes: bool(1), float64(2), int64(4), object(4)
memory usage: 27.8+ KB

Basic Statistics:
```

	Customer ID	Age	Total Spend	Items Purchased	Average Rating	Days Since Last Purchase
count	350.000000	350.000000	350.000000	350.000000	350.000000	350.000000
mean	275.500000	33.597143	845.381714	12.600000	4.019143	26.588571
std	101.180532	4.870882	362.058695	4.155984	0.580539	13.440813
min	101.000000	26.000000	410.800000	7.000000	3.000000	9.000000
25%	188.250000	30.000000	502.000000	9.000000	3.500000	15.000000
50%	275.500000	32.500000	775.200000	12.000000	4.100000	23.000000
75%	362.750000	37.000000	1160.600000	15.000000	4.500000	38.000000
max	450.000000	43.000000	1520.100000	21.000000	4.900000	63.000000

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Customer ID                 0
Gender                      0
Age                         0
```

```
City                        0
Membership Type             0
Total Spend                 0
Items Purchased             0
Average Rating              0
Discount Applied            0
Days Since Last Purchase    0
Satisfaction Level          0
dtype: int64
```

```
df["Satisfaction Level"].fillna(df["Satisfaction Level"].mode()[0], inplace=True)
```

```
/tmp/ipykernel_2583/4230081484.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are sett

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df


  df["Satisfaction Level"].fillna(df["Satisfaction Level"].mode()[0], inplace=True)
```

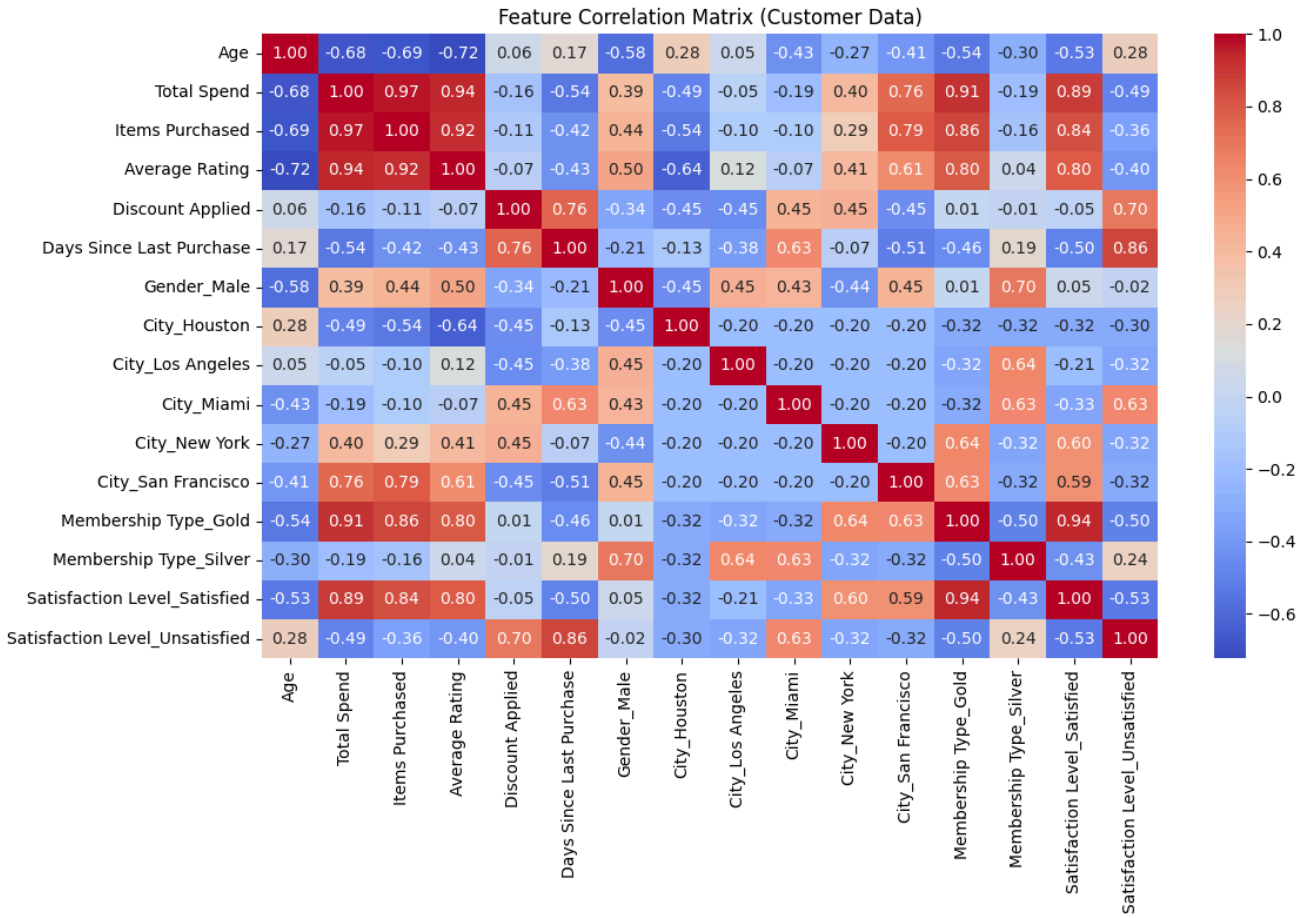
```
df = pd.get_dummies(df, drop_first=True)
```

```
df.info()
```

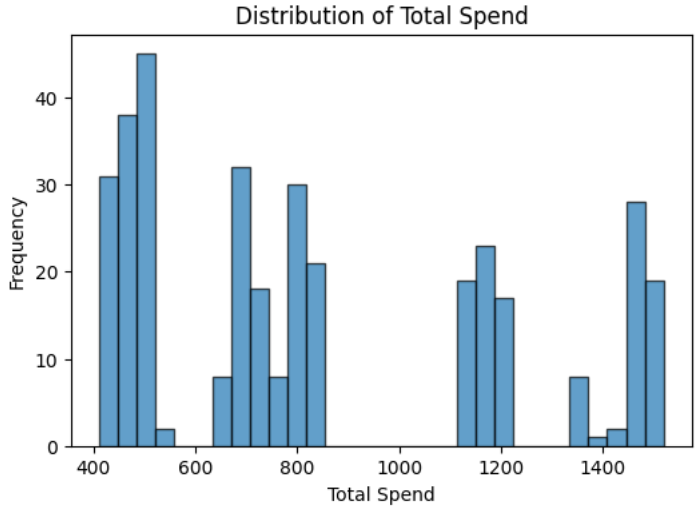
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 350 entries, 0 to 349
Data columns (total 17 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Customer ID                     350 non-null    int64
 1   Age                             350 non-null    int64
 2   Total Spend                     350 non-null    float64
 3   Items Purchased                 350 non-null    int64
 4   Average Rating                  350 non-null    float64
 5   Discount Applied                350 non-null    bool
 6   Days Since Last Purchase        350 non-null    int64
 7   Gender_Male                     350 non-null    bool
 8   City_Houston                    350 non-null    bool
 9   City_Los Angeles                350 non-null    bool
 10  City_Miami                      350 non-null    bool
 11  City_New York                   350 non-null    bool
 12  City_San Francisco              350 non-null    bool
 13  Membership Type_Gold            350 non-null    bool
 14  Membership Type_Silver          350 non-null    bool
 15  Satisfaction Level_Satisfied    350 non-null    bool
 16  Satisfaction Level_Unsatisfied  350 non-null    bool
dtypes: bool(11), float64(2), int64(4)
memory usage: 20.3 KB
```

```
df_corr = df.copy()
df_corr.drop("Customer ID", axis=1, inplace=True)
```

```
plt.figure(figsize=(12, 8))
correlation_matrix = df_corr.corr()
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Feature Correlation Matrix (Customer Data)")
plt.tight_layout()
plt.show()
```

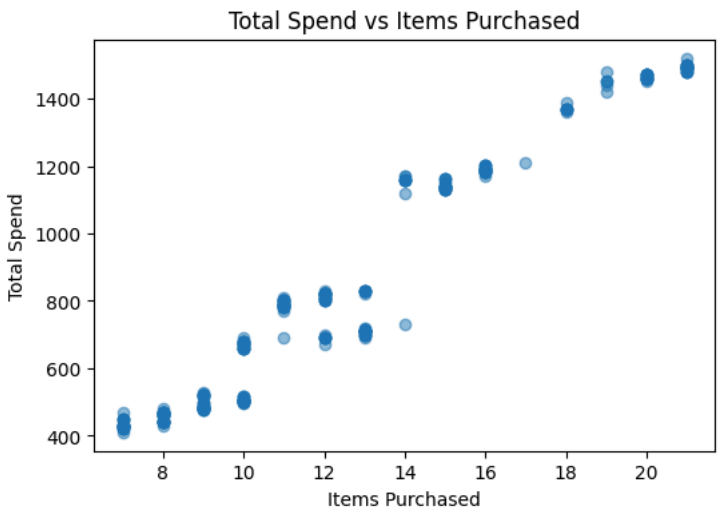



```
plt.figure(figsize=(6, 4))

plt.hist(df["Total Spend"], bins=30, edgecolor="black", alpha=0.7)

plt.title("Distribution of Total Spend")
plt.xlabel("Total Spend")
plt.ylabel("Frequency")

plt.show()
```



```
most_correlated = correlation_matrix["Total Spend"].abs().sort_values(ascending=False).index[1]
```

```
plt.figure(figsize=(6, 4))
plt.scatter(df_corr[most_correlated], df_corr["Total Spend"], alpha=0.5)
plt.title(f"Total Spend vs {most_correlated}")
plt.xlabel(most_correlated)
plt.ylabel("Total Spend")

plt.show()
```




```
target = "Total Spend"
```

```
correlation_matrix = df.corr(numeric_only=True)

most_correlated = correlation_matrix[target].abs().sort_values(ascending=False).index[1]
```

```
print("="*60)
print("NUMERICAL INSIGHTS FROM VISUALIZATIONS")
print("="*60)


print("\nTOTAL SPEND DISTRIBUTION INSIGHTS:")
print(f"Mean Spend: {df[target].mean():,.2f}")
print(f"Median Spend: {df[target].median():,.2f}")
print(f"Mode Spend (approx): {df[target].mode()[0]:,.2f}")
print(f"Std Dev: {df[target].std():,.2f}")
print(f"Skewness: {df[target].skew():.3f} (0=normal, >0=right-skewed)")
print(f"Spend Range: {df[target].min():,.2f} - {df[target].max():,.2f}")
```

```
============================================================
NUMERICAL INSIGHTS FROM VISUALIZATIONS
============================================================

TOTAL SPEND DISTRIBUTION INSIGHTS:
Mean Spend: 845.38
Median Spend: 775.20
Mode Spend (approx): 710.40
Std Dev: 362.06
Skewness: 0.563 (0=normal, >0=right-skewed)
Spend Range: 410.80 - 1,520.10
```

```
y = df['Total Spend']

X = df.drop(['Total Spend', 'Customer ID'], axis=1)
```

```


X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42
)

print(f"Training set size: {X_train.shape[0]} samples")
print(f"Test set size: {X_test.shape[0]} samples")
```

```
Training set size: 280 samples
Test set size: 70 samples
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
X_train_scaled = pd.DataFrame(X_train_scaled, columns=X_train.columns)
X_test_scaled = pd.DataFrame(X_test_scaled, columns=X_test.columns)
```

```
model = LinearRegression()
model.fit(X_train_scaled, y_train)
coefficients = pd.DataFrame({
    'Feature': X_train_scaled.columns,
    'Coefficient': model.coef_
}).sort_values('Coefficient', key=abs, ascending=False)

print("Model Coefficients (Feature Importance):")
print(coefficients)

print(f"\nIntercept: {model.intercept_:.4f}")
```

```
Model Coefficients (Feature Importance):
                            Feature  Coefficient
11            Membership Type_Gold   140.177998
1                  Items Purchased   100.839081
10              City_San Francisco    97.820878
9                    City_New York    80.977319
3                 Discount Applied   -48.484276
6                     City_Houston   -45.606393
12          Membership Type_Silver    33.681087
8                       City_Miami    27.110767
4         Days Since Last Purchase   -17.492091
7                 City_Los Angeles    15.687138
0                              Age     7.030793
5                      Gender_Male     5.877489
13    Satisfaction Level_Satisfied    -5.298696
2                   Average Rating     2.672836
14  Satisfaction Level_Unsatisfied    -1.752409

Intercept: 830.6375
```

```
y_train_pred = model.predict(X_train_scaled)
y_test_pred = model.predict(X_test_scaled)
```

```
comparison_df = pd.DataFrame({
    'Actual_Spend': y_test.values,
    'Predicted_Spend': y_test_pred,
    'Difference': (y_test_pred - y_test.values),
    'Absolute_Error': abs(y_test_pred - y_test.values)
})

comparison_df['Percentage_Error'] = abs(
    (comparison_df['Difference'] / comparison_df['Actual_Spend']) * 100
)

pd.options.display.float_format = '${:,.2f}'.format

print("All Test Set Predictions:")
print(comparison_df.head(20))

pd.reset_option('display.float_format')
```

```
All Test Set Predictions:
    Actual_Spend  Predicted_Spend  Difference  Absolute_Error  \
0        $810.90          $820.02       $9.12           $9.12
1      $1,140.60        $1,144.90       $4.30           $4.30
2        $710.40          $700.93      $-9.47           $9.47
3        $830.75          $828.84      $-1.91           $1.91
4        $430.80          $434.47       $3.67           $3.67
5        $470.50          $457.05     $-13.45          $13.45
6        $430.80          $426.64      $-4.16           $4.16
7      $1,140.60        $1,156.64      $16.04          $16.04
8      $1,450.50        $1,434.93     $-15.57          $15.57
9        $430.80          $434.47       $3.67           $3.67
10     $1,490.10        $1,486.47      $-3.63           $3.63
11       $690.60          $692.34       $1.74           $1.74
12     $1,160.30        $1,139.58     $-20.72          $20.72
```

```
13     $1,160.30        $1,139.58     $-20.72          $20.72
14       $430.80          $434.47       $3.67           $3.67
15     $1,460.50        $1,460.38      $-0.12           $0.12
16       $660.30          $663.00       $2.70           $2.70
17       $470.50          $462.27      $-8.23           $8.23
18       $485.25          $491.61       $6.36           $6.36
19       $670.30          $669.52      $-0.78           $0.78

    Percentage_Error
0              $1.13
1              $0.38
2              $1.33
3              $0.23
4              $0.85
5              $2.86
6              $0.97
7              $1.41
8              $1.07
9              $0.85
10             $0.24
11             $0.25
12             $1.79
13             $1.79
14             $0.85
15             $0.01
16             $0.41
17             $1.75
18             $1.31
19             $0.12
```

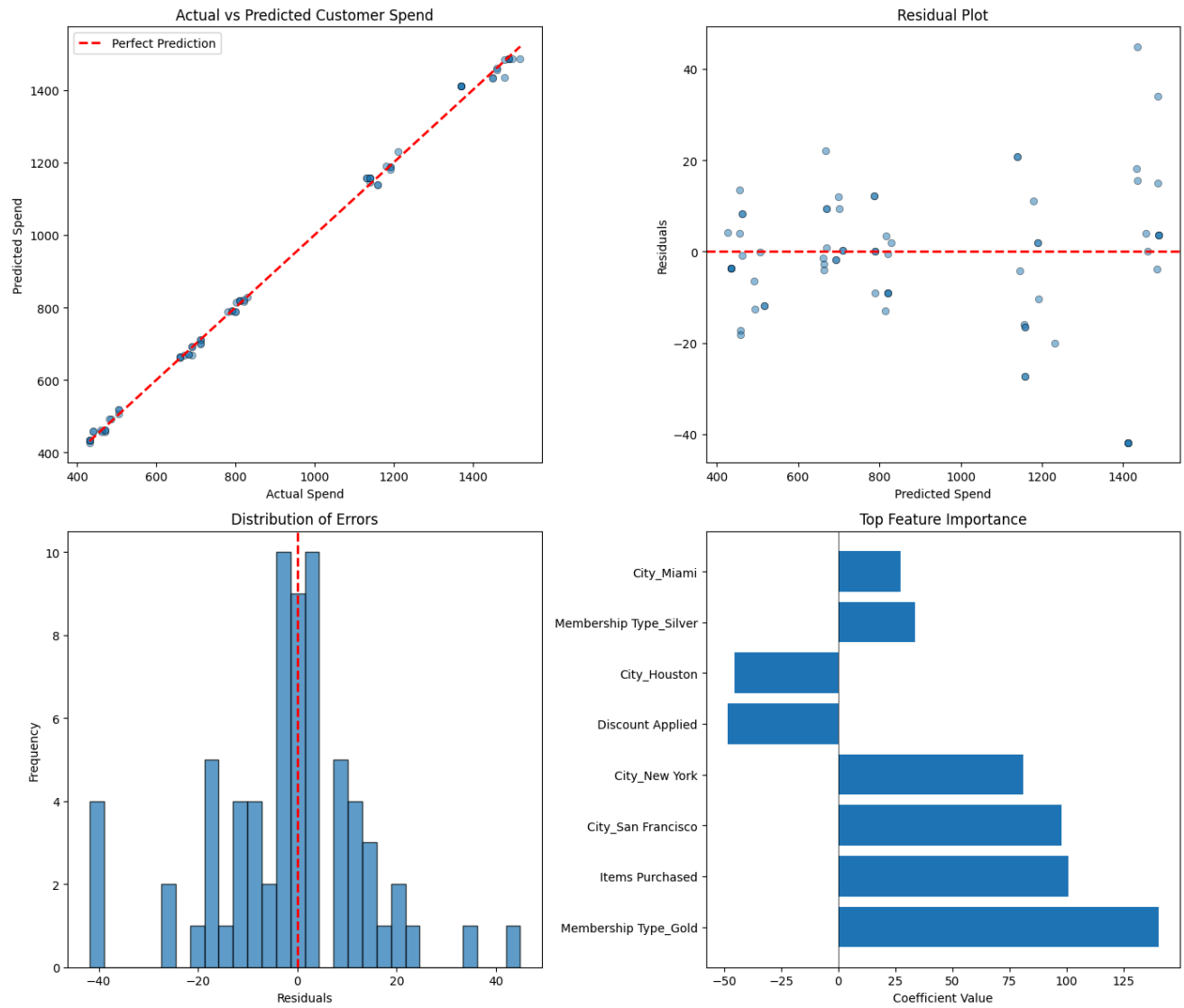
```
detailed_predictions = X_test.copy()

detailed_predictions['Actual_Spend'] = y_test.values
detailed_predictions['Predicted_Spend'] = y_test_pred
detailed_predictions['Difference'] = y_test_pred - y_test.values
detailed_predictions['Percent_Error'] = abs(
    (detailed_predictions['Difference'] / detailed_predictions['Actual_Spend']) * 100
)

print("\nDetailed Predictions with Features:")
print(
    detailed_predictions[
        list(X_test.columns) +
        ['Actual_Spend', 'Predicted_Spend', 'Difference', 'Percent_Error']
    ].head(10)
)
```

```
157                       13         True         False              True
341                       36        False         False             False
315                       61         True         False             False
234                       14         True         False              True
155                       23        False          True             False
274                       22        False          True             False
304                       29        False          True             False
227                       27        False         False             False
278                       12         True         False             False
185                       23        False          True             False

     City_Miami  City_New York  City_San Francisco  Membership Type_Gold  \
157       False          False               False                 False
341       False           True               False                  True
315        True          False               False                 False
234       False          False               False                 False
155       False          False               False                 False
274       False          False               False                 False
304       False          False               False                 False
227       False           True               False                  True
```

```
     Satisfaction_Level_Unsatisfied  Actual_Spend  Predicted_Spend  \
157                           False        810.90       820.023075
341                           False       1140.60      1144.895804
315                            True        710.40       700.933464
234                           False        830.75       828.839443
155                           False        430.80       434.465529
274                           False        470.50       457.051608
304                           False        430.80       426.638409
227                           False       1140.60      1156.636484
278                           False       1450.50      1434.930271
185                           False        430.80       434.465529

     Difference  Percent_Error
157    9.123075       1.125056
341    4.295804       0.376627
315   -9.466536       1.332564
234   -1.910557       0.229980
155    3.665529       0.850866
274  -13.448392       2.858319
304   -4.161591       0.966015
227   16.036484       1.405969
278  -15.569729       1.073404
185    3.665529       0.850866
```

```
def evaluate_model(y_true, y_pred, dataset_name):
    """Calculate and print evaluation metrics"""

    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)
    mae = mean_absolute_error(y_true, y_pred)
    r2 = r2_score(y_true, y_pred)

    print(f"\n{dataset_name} Performance:")
    print(f"MSE:  {mse:.4f}")
    print(f"RMSE: {rmse:.4f}")
    print(f"MAE:  {mae:.4f}")
    print(f"R²:   {r2:.4f}")

    return {
        'MSE': mse,
        'RMSE': rmse,
        'MAE': mae,
        'R2': r2
    }


train_metrics = evaluate_model(y_train, y_train_pred, "Training")
test_metrics = evaluate_model(y_test, y_test_pred, "Test")
```

```
Training Performance:
MSE:  193.3068
RMSE: 13.9035
MAE:  10.5880
R²:   0.9985

Test Performance:
MSE:  258.9506
RMSE: 16.0919
MAE:  11.3702
R²:   0.9981
```

```
fig, axes = plt.subplots(2, 2, figsize=(14, 12))

axes[0, 0].scatter(y_test, y_test_pred, alpha=0.5, edgecolors='k', linewidth=0.5)
axes[0, 0].plot([y_test.min(), y_test.max()],
                [y_test.min(), y_test.max()],
                'r--', lw=2, label='Perfect Prediction')

axes[0, 0].set_xlabel('Actual Spend')
axes[0, 0].set_ylabel('Predicted Spend')
axes[0, 0].set_title('Actual vs Predicted Customer Spend')
axes[0, 0].legend()

residuals = y_test - y_test_pred

axes[0, 1].scatter(y_test_pred, residuals, alpha=0.5, edgecolors='k', linewidth=0.5)
axes[0, 1].axhline(y=0, color='r', linestyle='--', linewidth=2)
```

```
axes[0, 1].set_xlabel('Predicted Spend')
axes[0, 1].set_ylabel('Residuals')
axes[0, 1].set_title('Residual Plot')

axes[1, 0].hist(residuals, bins=30, edgecolor='black', alpha=0.7)
axes[1, 0].set_xlabel('Residuals')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].set_title('Distribution of Errors')
axes[1, 0].axvline(x=0, color='r', linestyle='--', linewidth=2)

coeff_plot = coefficients.head(8)

axes[1, 1].barh(coeff_plot['Feature'], coeff_plot['Coefficient'])
axes[1, 1].set_xlabel('Coefficient Value')
axes[1, 1].set_title('Top Feature Importance')
axes[1, 1].axvline(x=0, color='black', linestyle='-', linewidth=0.5)

plt.tight_layout()
plt.show()
```